CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Theory Examination (IT)

November 2012

IT-304: Modeling Simulation and Operational Research

Date: 30.11.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Attempt following. **[07]**

1. In Kendell's notations {(a/b/c) : (d/e)}, 'd' stands for_____.
2. Customer behavior in which he moves from one queue to another in a multiple channel situation is _____.
3. _____ Method is used to simulate randomly generated data.
4. Explain Pure Birth process.
5. Define Critical Path.
6. Dijkstra's algorithm is used to _____.
7. PERT stands for _____.

Q - 2 (a) A project consists of 5 activities A to E. The completion time for each activity is a random variable. The data considering probability distribution along with completion time for each activity is as follows: **[08]**

Activity	Immediate Predecessor	Time (Day)/ Prob. 1	2	3	4
A	-	0.2	0.3	0.5	---
B	A	---	0.4	0.2	0.4
C	A	0.1	---	0.8	0.1
D	B	0.2	0.3	0.4	0.1
E	C, D	0.9	0.1	---	---

Activity	Run 1	Run 2
A	12	93
B	14	08
C	39	12
D	22	52
E	67	49

Using above information and 2 runs of simulation, simulate average project completion time. Use random numbers given in above table.

(b) Consider the Queuing Model {(M/M/1) : (∞/FCFS)}. List out formulae to find length of queue, length of system, waiting time in queue, waiting time in system when inter arrival rate and service rate is given. **[06]**

OR

Q - 2 (a) A television repairman finds that the time spent on his job has an exponential distribution with a mean of 30 minutes. If he repairs sets in the order in which they came in, and if the arrival of sets follows a Poisson distribution approximately with an average rate of 10 per 8-hour day, what is the repairman's expected idle time each day? Find out traffic intensity. How many jobs are ahead of the average set just brought in? **[08]**

(b) Given **Figure 1 [Page 3]** represents permissible route and distances in miles between city 1 and other cities from 2 to 5. Determine shortest route from 1 to 5. **[06]**

Q - 3 (a) A dentist schedules all his patients for 30 – minute appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and time actually needed to complete the work: **[08]**

Simulate the dentist's clinic for two hours and determine the average waiting time for the patients as well as the idleness of the doctor. Assume that all the patients show up

at the clinic at exactly their scheduled arrival time starting at 9:00 a.m. Use the following random numbers for handling the above problem: 40, 82, 11, 34, 25, 66,17

Category of service	Time required (mins)	Probability of category
Filling	45	0.40
Crowning	60	0.15
Cleaning	15	0.15
Extraction	45	0.10
Check – up	15	0.20

(b)	Define Spanning tree. Apply Prim's algorithm to find out minimal spanning tree for the graph given in **Figure 2[Page 3]**.	[06]
	OR	
Q - 3 (a)	Apply Floyd's algorithm on **Figure 2 [Page 3]** to find shortest route between any two nodes. Also discuss about shortest route between vertices 3 and 4 with its length.	[08]
(b)	Consider a single server queuing system with Poisson input, exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hours and the maximum permissible calling units in the system is two. Derive the steady – state probability distribution of the number of calling units in the system, and then calculate the expected number in the system.	[06]
	SECTION – II	
Q - 4 (a)	Discuss the Application of Operations Research in engineering.	[03]
(b)	Use Graphical method solve the linear programming problem: $Max\ Z = 15x_1 + 10x_2$; $sub\ to\ 4x_1 + 6x_2 \le 360;\quad 3x_1 \le 180;\quad 5x_2 \le 200$, where $x_1, x_2 \ge 0$	[04]
Q - 5 (a)	1. **True** or **False** "For a maximization problem in Big-M method, the objective function coefficient for an artificial variable is M." 2. **True** or **False** "In the optimal simplex table z_j-c_j=0 value indicates existence of an alternative solution of the problem." 3. Every _____ Problem is a Special case of Transportation Problem. 4. Graphical method to solve LPP is applicable only for ____ number of variables. 5. Define feasible solution.	[01] [01] [01] [01] [02]
(b)	Solve the following LPP by using simplex method. $Max\ Z = 3x_1 + 5x_2$; $sub\ to\ 3x_1 + 2x_2 \le 18;\quad x_1 \le 4;\quad x_2 \le 6$, where $x_1, x_2 \ge 0$	[08]
	OR	
Q - 5 (a)	1. **True** or **False** "A basic feasible solution is called degenerate if values of all basic variables are non-zero and positive." 2. **True** or **False** "An optimum basic feasible solution is always a basic feasible solution."	[01] [01]
(b)	Solve the following LPP by using Big-M method. Maximize $Z = 5X_1 + X_2$ subject to $5X_1 + 2X_2 \le 20;\ X_1 \ge 3;\ X_2 \le 5$ where, $X_1,\ X_2 \ge 0$	[08]
(c)	GNFC chemicals manufactures two products say, naphtha (x) and urea (y) in its plant located near by Baroda. It gets a profit of Rs. 70 per unit for naphtha and Rs. 100 per unit for urea. The time requirements for each product and the total available time in each department are shown in the following **table**. Each product has to pass through two departments in the plant. Further, the demand for these products restricts the production to a maximum of 450	[04]

units of each product. Express this LPP mathematically. [Please refer table]

Department	Time required per unit (in hours)		Hours available in a month
	Naphtha	Urea	
1	3	4	2100
2	4	3	2100

Q - 6 (a)

1. **True** or **False** "VAM always gives optimal Solution to any TP." **[01]**
2. **True** or **False** "The solution to a TP with m-rows and n-columns is feasible if number of positive allocations is (m+n-1)." **[01]**

(b) Match Pairs **[04]**

Primal	Dual
1) Objective function is of minimization	a) constraint j is = type
2) constraint i is = type	b) $\geq$ type constraints
3) $x_j \geq 0$ variables	c) i^{th} dual variable, $y_i \geq 0$
4) Variable x_j unrestricted in sign	d) Variable y_i unrestricted in sign
	e) Objective function is of maximization

(c) Find initial solution of the following Transportation Problem using NWCM, LCM and VAM. Also compare the results obtained by each method as conclusion. **[08]**

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

OR

Q - 6 (a)

1. **True** or **False** "The dummy source or destination in a TP is added to dissatisfy rim conditions." **[01]**
2. **True** or **False** "One disadvantage of using NWCM to find initial solution to the TP is that it does not take into account cost of transportation." **[01]**

(b) Convert the given Primal problem in to a dual. **[04]**

$Max\, Z = 3x_1 + 5x_2;\ \ sub\ to\ 3x_1 + 2x_2 \leq 18;\quad x_1 \leq 4;\quad x_2 \leq 6$, where $x_1, x_2 \geq 0$

(c) Consider the problem of assigning 5 jobs to 5 persons. The assignment costs (in Rs.) are given as follows. Determine the optimal assignment schedule by using Hungarian method of assignment and also give the minimum total assignment cost. **[08]**

		Jobs				
		I	II	III	IV	V
PERSONS	A	8	4	2	6	7
	B	0	9	5	5	4
	C	3	8	9	2	6
	D	4	3	1	0	3
	E	9	5	8	9	5

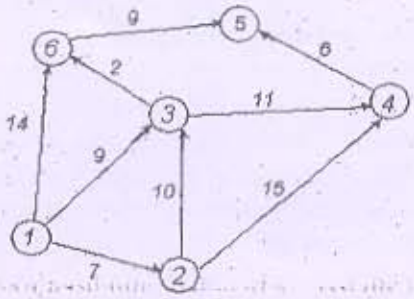

Figure 1

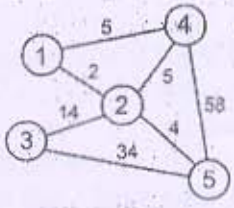

Figure 2